



Chapter 1

Introduction

1.1 Basic Concepts

- 1.1.1 What is an Image?
 - **photograph** == Image?

Sample of Photograph

- Human being
 - Lena: most famous picture in the field of Digital Image Processing



Sample of Photographs

- Landscape



Sample of Photographs

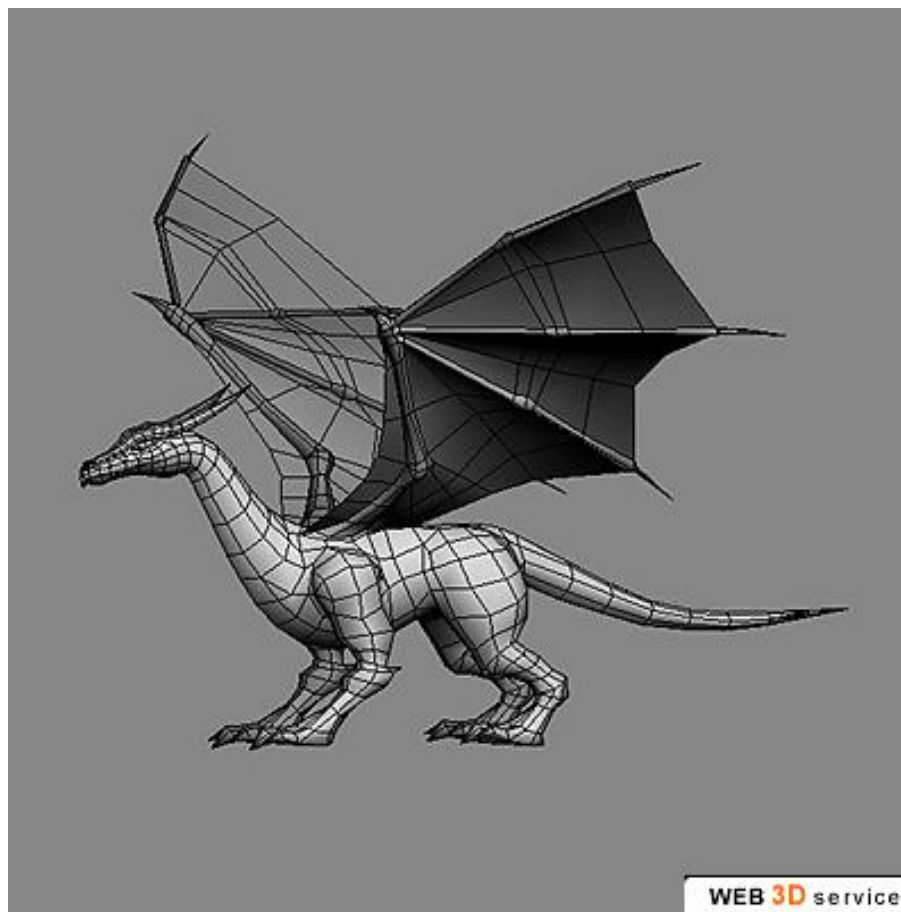
- Flower



An oil painting is an image, but is NOT a photograph



Am Image of 3D Model



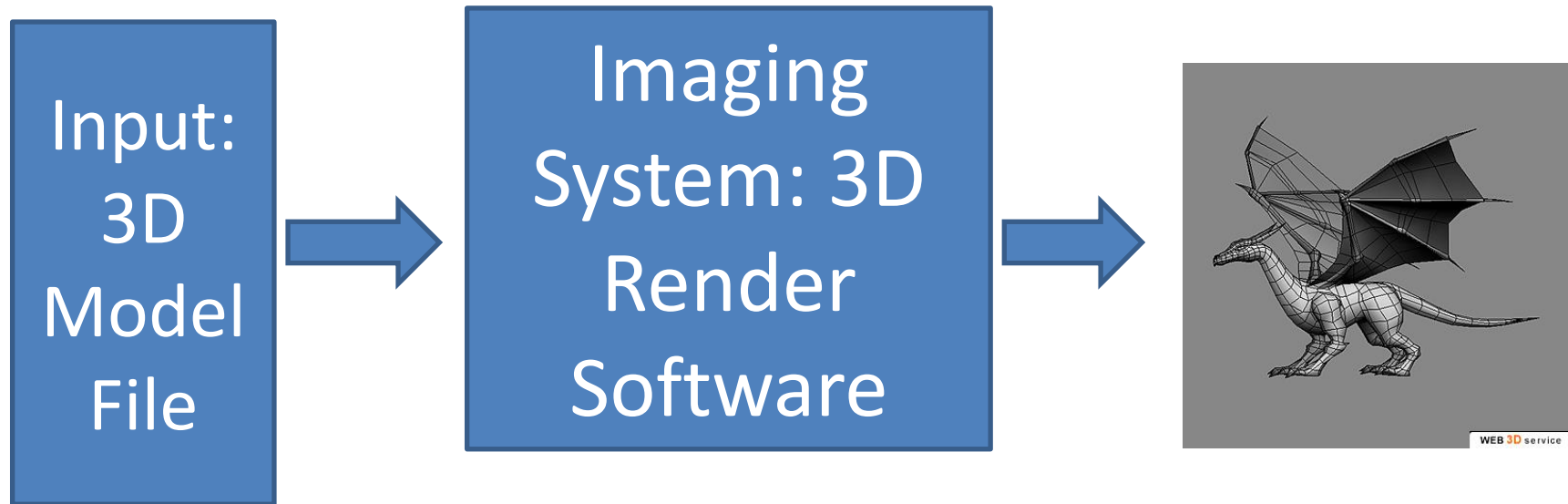
What is an Image



What is a Photograph?



Image of 3D Model



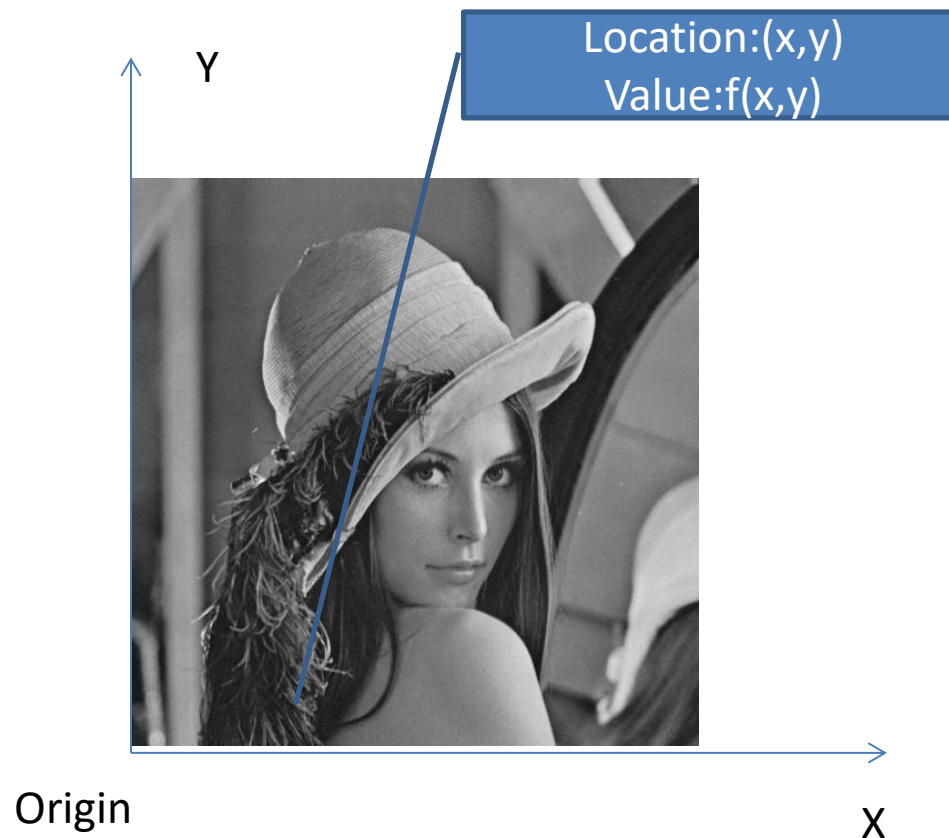
Color Image and gray level image



1.1.1 What is an Image? (Gray Level)

- An image can be define as a two-dimensional function $f(x, y)$, where x and y are spatial coordinates, and the value of amplitude or value of f at (x, y) is call the intensity or gray level of f at that point.

Mathematical model of Image



What is an Color Image

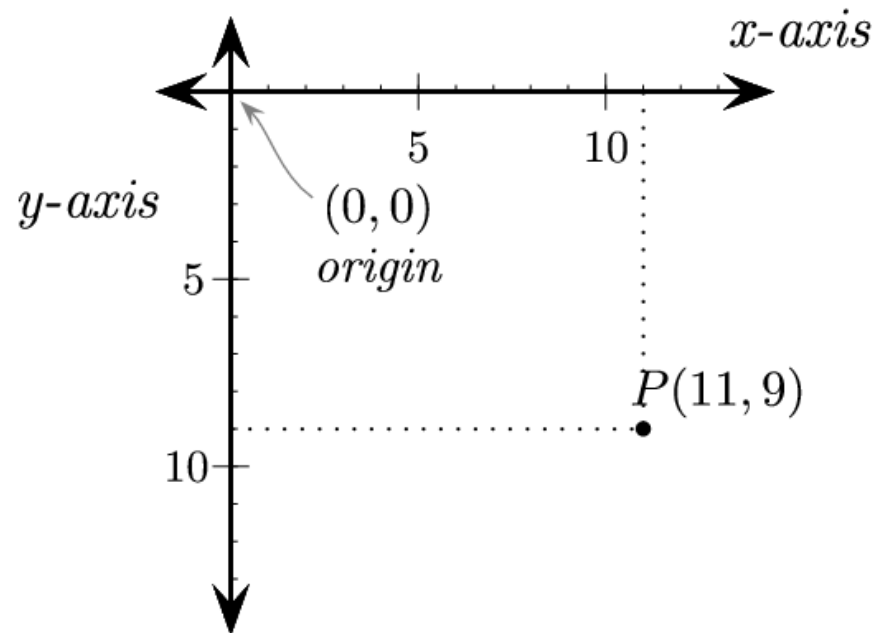
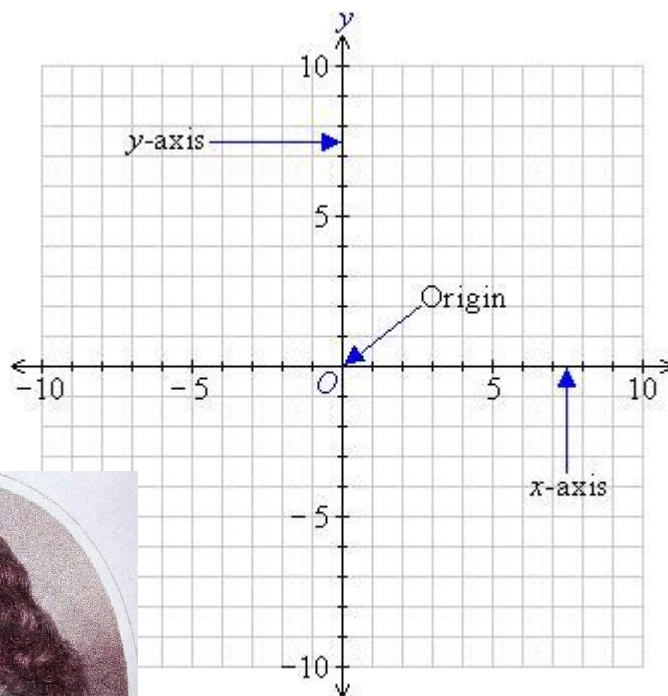
- $f(x,y)$ is a vector, for example, $[r,g,b]^T$

1.1.2 What is a Digital Image

- When x , y , intensity of f are all finite, discrete quantities, we call the image as Digital Image.
- Because the coordinate (x, y) are discrete, so a digital image can be considered as a Matrix or a 2D array.

Carte'sian coordinate system

Graphics Coordinate System



1.1.2 What is a Digital Image

- Gray level digital image: 2-D byte array
- Range of byte: [min max]
 - Min: Black
 - Max: White

[Programming of “Hello, Image”]

[Introduction to C--]

1.1.3 Pixel

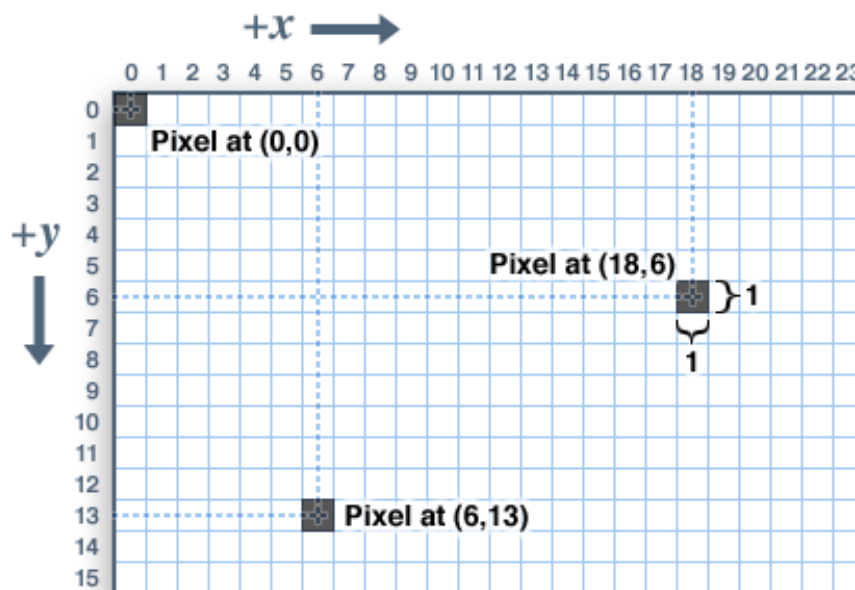
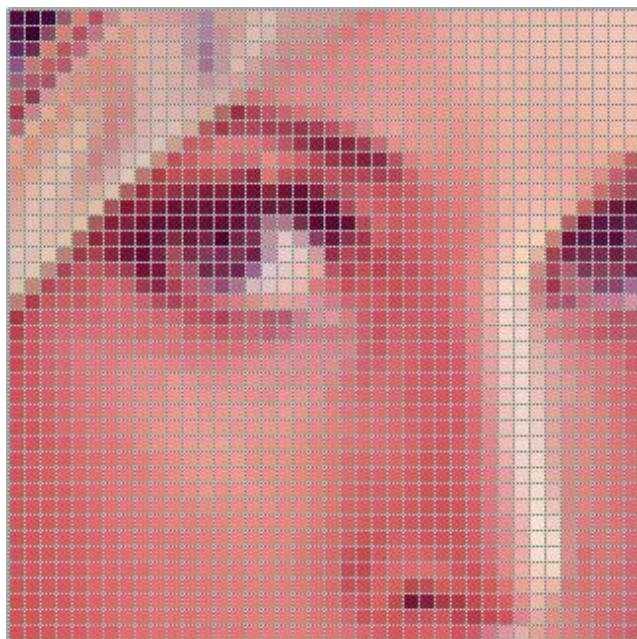
- Pixel can be see the element of picture or image.
- Each pixel has a particular location ((x, y) coordinate) and value(intensity or color)
- [Programming: How to access pixel, read and write operation]

gray	gray	gray	gray
gray	gray	black	black
gray	gray	black	black
gray	gray	gray	gray

1.1.3 Pixel

A0	A0	A0	A0	A0	A0	A0	A0
A0	A0	A0	A0	A0	A0	A0	A0
A0	A0	0	0	0	0	A0	A0
A0	A0	0	0	0	0	A0	A0
A0	A0	0	0	0	0	A0	A0
A0	A0	0	0	0	0	A0	A0
A0	A0	A0	A0	A0	A0	A0	A0
A0	A0	A0	A0	A0	A0	A0	A0

1.1.3 Pixel



1.1.4 Image Resolution

- Logic Resolution, Image Size:
 - How many pixels in a image

Camera Sensor Resolution

- Resolution: 2 megabyte pixels



Screen Resolution

- VGA resolution: 640 X 480
- HD resolution: 1920 X 1080

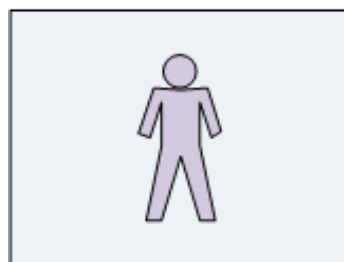


Let's Go Digital

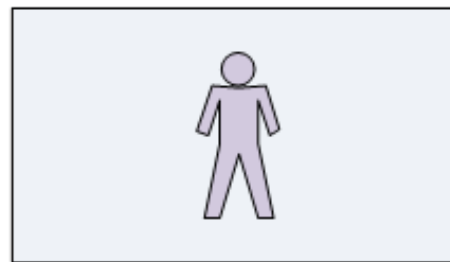
Aspect Ratio

- Aspect Ratio = width / height
- 2 megabyte pixels = 1600X1200

Picture Aspect Ratio



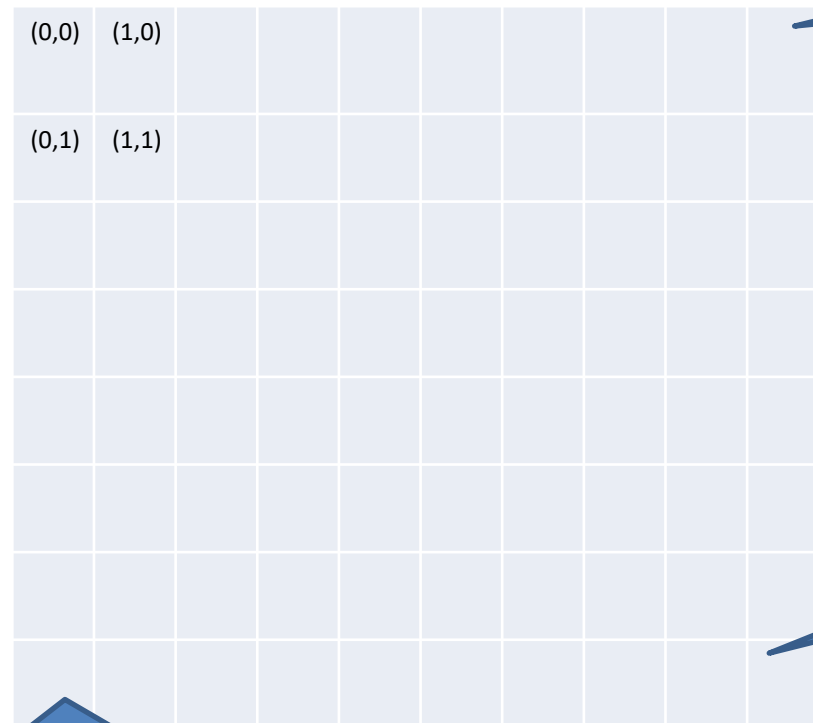
4 : 3



16 : 9

1.1.4 Image Resolution

- [Programming: Image Size]



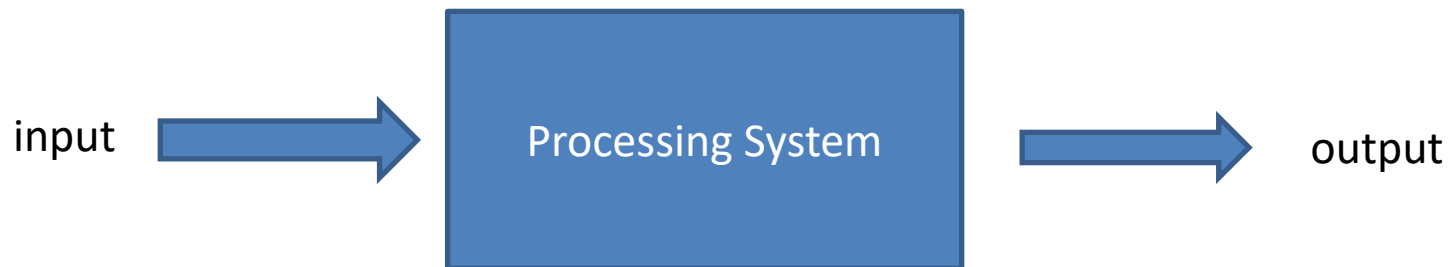
$(f.GetLength(0)-1, 0)$

$(f.GetLength(0)-1, f.GetLength(1)-1)$

$(0, f.GetLength(1)-1)$

1.2 What is Digital Image Processing

- Digital image processing means processing digital images by means of digital computers.



1.3 Applications of Image Processing

- 1.3.1 Low Level Image Processing
 - Spatial Transforms
 - [Demo of same picture view software]

Image Enhancement

- To get a better image

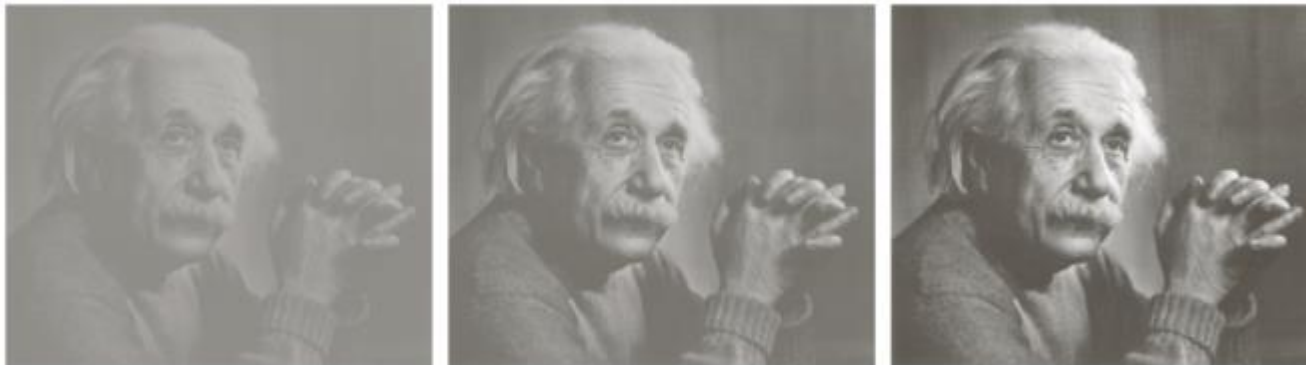
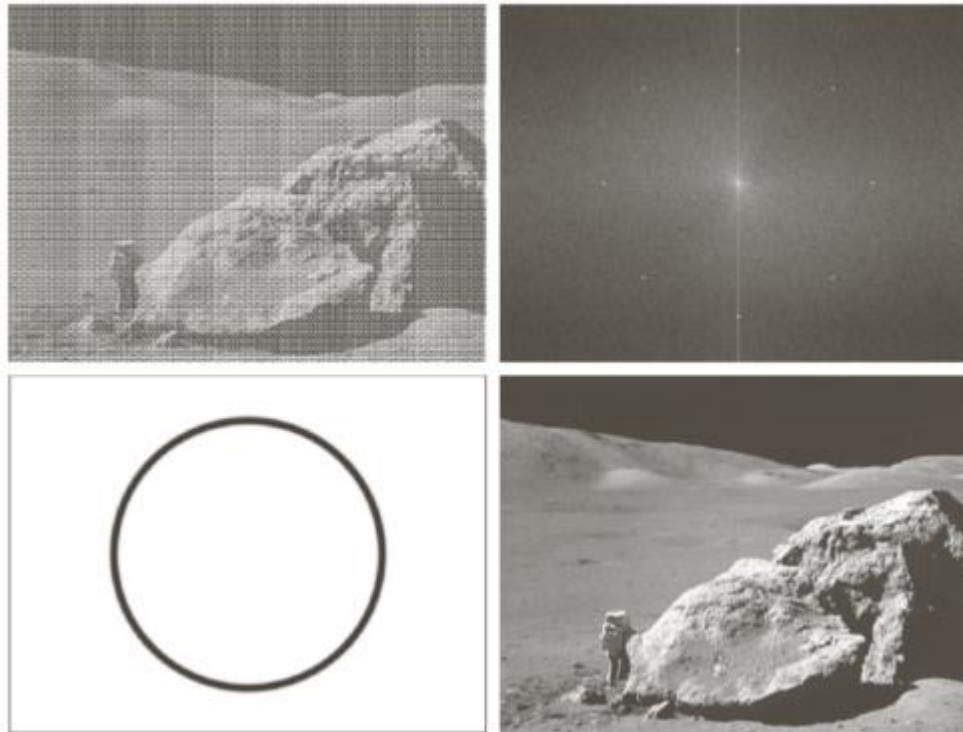


Image Restoration

- To get more “real” image



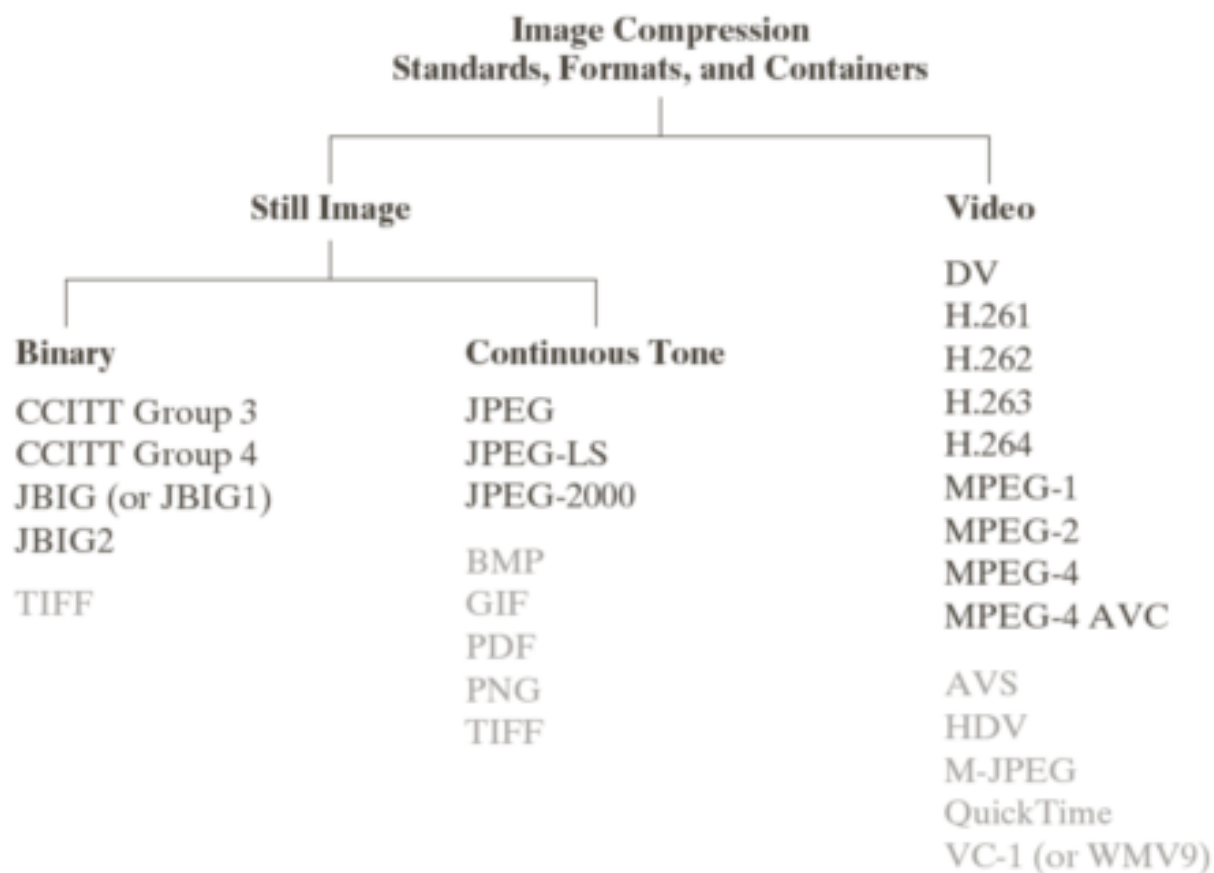
1.3.2 Image Compression

- Why we need this Technology
 - The amount data of image or video may be extremely huge

1.3.2 Image Compression

- The amount data of 2 hour of SD Video.
- SD means Standard Definition, resolution is 720X480.
- To storage a pixel of color image need 3bytes.
- One second data is
 $720 \times 480 \text{ (resolution)} \times 3 \times 30 \text{ (frames/sec)} = 31,104,000 \text{ bytes}$
- Total amount is
 $31,104,000 \times 3600 \text{ (sec/h)} \times 2 = 2.24 \times 10^8 \text{ bytes}$
or 224 GB (gigabyte)

1.3.2 Image Compression

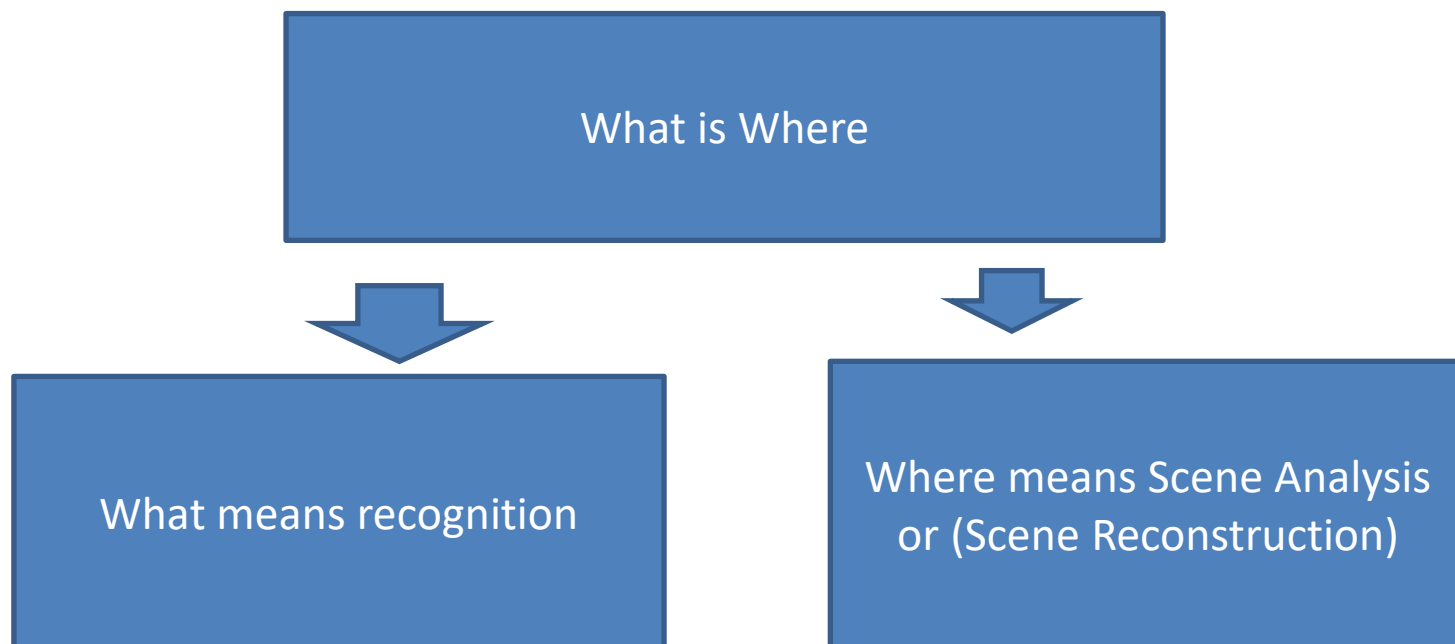


1.3.2 Image Compression

- Data Compress: **entropy coding, lossless compression**
- Image Compress : **DCT, entropy coding, ext, lossy compression**

1.3.3 computer vision or machine vision

Use computer and image sensor to emulate human vision



1.3.3 computer vision (Microsoft)



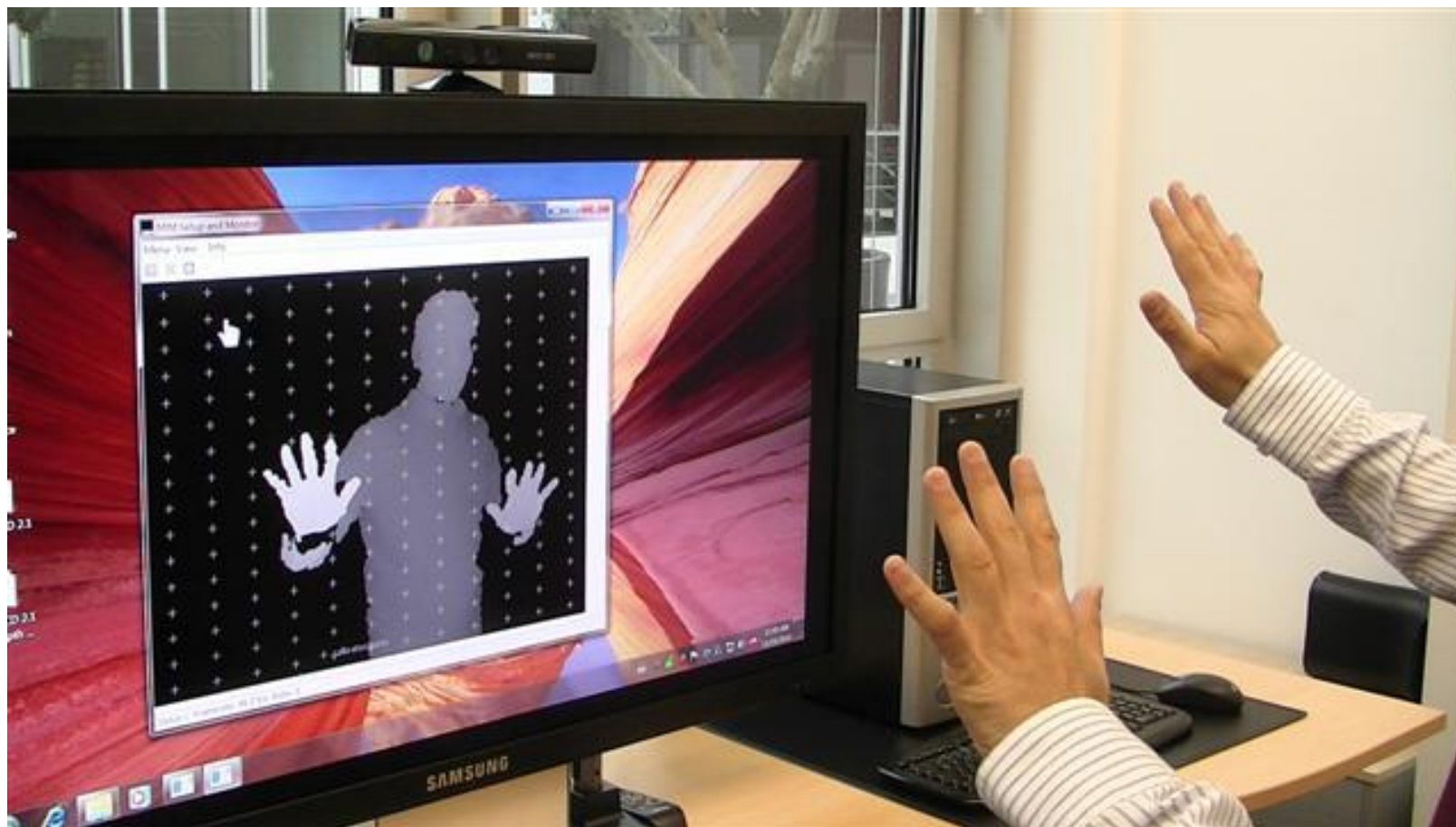
computer vision (RGBD Camera)



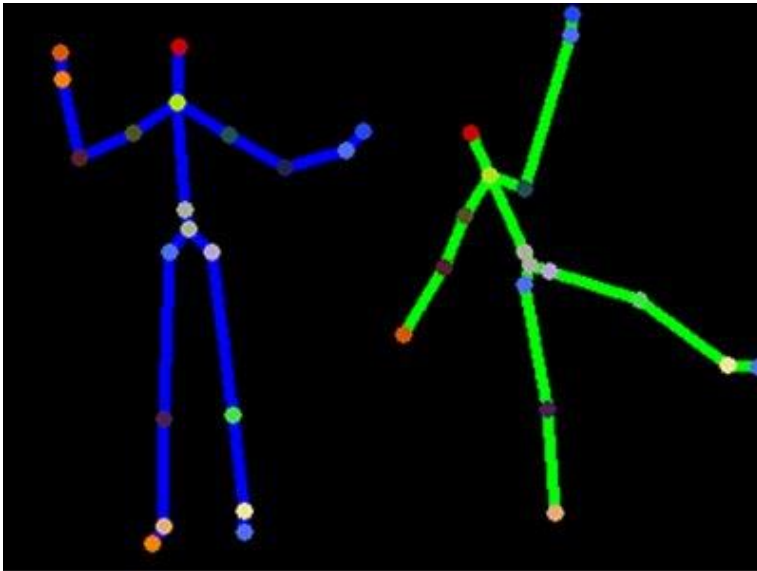
KINECT™
for XBOX 360.



computer vision (Depth Image)



computer vision (Pose Estimation)



Computer Vision

- **Low-level Processing: both input and output are images**
- **Mid-level Processing: input images, output attributes of those images**
 - Image Segmentation Image Indexing (Feature Extraction)
- **High-level Processing: related to computer vision**
 - Image Analysis and Understanding